

Introduction to HTML and CSS - HTML Lab

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Topics Covered

- What is HTML?
- HTML Syntax
- Tags:
 - html, head, title, meta, body,
 - h1, h2, h3, h4, h5, h6, p, a, strong, em, ul, ol, li, dl, dt, dd, img, blockquote
 - main, aside, article, section, header, footer
 - details, summary, figure, caption

Resources

- HTML Element references on MDN:
 - <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- HTML Basics
 - https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/HTML_basics
- Semantics:

- <https://developer.mozilla.org/en-US/docs/Glossary/Semantics>
- **W3 Validator:**
 - <https://validator.w3.org>

Tasks

1. Simple HTML Document

Create an HTML Document and follow the requirements listed below to complete the task.

Heading level 1 Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Heading level 2 Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Heading level 3 Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Heading level 4 Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Heading level 5 Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Heading level 6 Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci. Lorem ipsum dolor sit amet, consectetur adipisicing elit. Eius porro assumenda ex, cum beatae aspernatur explicabo nulla impedit ab, odio ea magni unde eum dicta? Eligendi debitis dolorem eius adipisci.

Requirements

- Use the copy provided in the text file.
- Create an HTML Document with **html, head, title, body**.
- Don't forget the Doctype declaration.
- Body needs to contain at **6 title – h1, h2, h3, h4, h5, h6**.
- Body needs to contain at least **12 paragraphs**.
- Place **2 paragraphs** after each title.

Resources

- <https://developer.mozilla.org/en-US/docs/Glossary/Doctype>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/html>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/head>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/title>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/body>
- https://developer.mozilla.org/en-US/docs/Web/HTML/Element/Heading_Elements
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/p>

2. Ordered & Unordered Lists

Create an HTML Document and follow the requirements listed below to complete the task.

Below you can see an unordered list

- List item 1
- List item 2
- List item 3
- List item 4

Below you can see an ordered list

1. List item 1
2. List item 2
3. List item 3
4. List item 4

Requirements

- Use the copy provided in the text file.
- Body needs to contain **1 ul** elements with **4 li** elements each.
- Body needs to contain **1 ol** element with **4 li** elements each.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ul>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ol>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/li>

3. Nested Ordered & Unordered Lists

Create an HTML Document and follow the requirements listed below to complete the task.

Below you can see an unordered list

- List item 1
- List item 2
 - List item 2 1
 - List item 2 3 1
 - List item 2 3 2
 - List item 2 3 3
 - List item 2 3 4
 - List item 2 2
 - List item 2 3
 - List item 2 4
- List item 3
- List item 4

Below you can see an ordered list

1. List item 1
2. List item 2
 1. List item 2 1
 2. List item 2 2
 3. List item 2 3
 1. List item 2 3 1
 2. List item 2 3 2
 3. List item 2 3 3
 4. List item 2 3 4
 4. List item 2 4
3. List item 3
4. List item 4

Requirements

- Use the copy provided in the text file.
- Body needs to contain **3 ul** elements with **4 li** elements each.
- **ul** elements should be nested in each other.
- The **second li** elements of the **first-level ol** should contain a **ol**.
- The **third li** elements of the **second-level ol** should contain a **ol**.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ul>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/ol>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/li>

4. Definition List

Create an HTML Document and follow the requirements listed below to complete the task.

HTML
HyperText Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser. It defines the content and structure of web content. It is often assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript.

CSS
Cascading Style Sheets (CSS) is a style sheet language used for specifying the presentation and styling of a document written in a markup language such as HTML or XML (including XML dialects such as SVG, MathML or XHTML). CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript.

JavaScript
Programming language and core technology of the World Wide Web, alongside HTML and CSS. As of 2024, 98.8% of websites use JavaScript on the client side for webpage behavior, often incorporating third-party libraries. All major web browsers have a dedicated JavaScript engine to execute the code on users' devices.

Requirements

- Use the copy provided in the text file.
- Body needs to contain **1 dl** element, **3 dt** elements and **3 dd** elements.
- **dt** and **dd** elements should be inside the **dl** element.
- **dd** elements should follow a **dt** element.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/dl>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/dt>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/dd>

5. Marking Text – Links, Strong, Emphasis and more

Create an HTML Document and follow the requirements listed below to complete the task.

Separation of concerns

In [computer science](#), **separation of concerns** is a *design principle* for separating a computer program into distinct sections. Each section addresses a *separate concern*, a set of information that affects the code of a computer program. A concern can be as general as "the details of the hardware for an application". A program that embodies SoC well is called a modular program. Modularity, and hence separation of concerns, is achieved by encapsulating information inside a section of code that has a well-defined interface. Encapsulation is a means of information hiding. Layered designs in information systems are another embodiment of separation of concerns (e.g., presentation layer, business logic layer, data access layer, persistence layer).

Separation of concerns results in more degrees of freedom for some aspect of the program's design, deployment, or usage. Common among these is increased freedom for simplification and maintenance of code. When concerns are well-separated, there are more opportunities for module upgrade, reuse, and independent development. Hiding the implementation details of modules behind an interface enables improving or modifying a single concern's section of code without having to know the details of other sections and without having to make corresponding changes to those other sections. Modules can also expose different versions of an interface, which increases the freedom to upgrade a complex system in piecemeal fashion without interim loss of functionality.

Separation of concerns is a form of abstraction. As with most abstractions, separating concerns means adding additional code interfaces, generally creating more code to be executed. The extra code can result in higher computation costs in some cases, but in other cases also can lead to reuse of more optimized code. So despite the many benefits of well-separated concerns, there may be an associated execution penalty - though with today's powerful hardware that penalty may in most practical cases be negligible.

Requirements

- Use the copy provided in the file.
- Body element must contain:
 - o 1 **a** element
 - o 1 **strong** element
 - o 1 **em** element
 - o 1 **abbr** element

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/a>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/strong>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/em>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/cite>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/sup>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/abbr>

6. Images

Create an HTML Document and follow the requirements listed below to complete the task.

Images

The `img` element embeds an image into the document.

Lorem ipsum dolor, sit amet consectetur adipiscing elit. Modi distinctio asperiores accusamus dolorum dolores aliquam vero iusto velit? Illo quas beatae, soluta amet ea id repellat iure aspernatur recusandae nulla!



Minus tenetur quod, dolor, hic ratione quidem quo accusamus quos vitae numquam asperiores, officiis incidunt aliquam. Sed modi quae reiciendis quam ratione?



Sit quas dolorum vel ad fugiat nulla, esse at numquam libero reiciendis saepe rerum voluptatem? Inventore molestias natus esse, doloribus sint recusandae?

Images next to each other



Image Gallery (List of images)



Requirements

1. **File Setup:**
 - Create an **HTML** document named **images.html**
2. **HTML Requirements**
 - Use **img** tag and get different images from **unsplash.com**
 - Make sure you add one **ul** with **li** items and images inside

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/img>

7. Semantic Tags – Header, Main, Footer

Create an HTML Document and follow the requirements listed below to complete the task.

This is the header

This is the main area of the site

This is the footer.

Requirements

1. **File Setup:**
 - Create an **HTML** document named **semantic-tags.html**
2. **HTML Requirements**
 - Create the simple document structure using the following tags:
 - **header**
 - **main**
 - **footer**

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/header>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/footer>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/main>

8. Semantic Tags – Article, Section

Create an HTML Document and follow the requirements listed below to complete the task.

Article title goes here

Section title

Lorem ipsum dolor sit amet consectetur adipiscing elit. Facilis distinctio veniam quidem velit mollitia et optio debitis molestiae illum. Incidunt sapiente, praesentium numquam perferendis deleniti modi assumenda quod eveniet similique!

Published on 12.03.2024 at 15:00 by David Core

Requirements

1. **File Setup:**
 - Create an **HTML** document named **semantic-tags-part2.html**

2. HTML Requirements

- Create the simple document structure using the following tags:
 - **article**
 - **section**

Hint: Never place the main title (**<h1>**) in **<article>** and **<section>** elements, as a web page can contain several of them and the main title should be unique.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/article>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/section>

9. Semantic Tags – Details, Summary

Create an HTML Document and follow the requirements listed below to complete the task.



Requirements

1. **File Setup:**
 - Create an **HTML** document named **semantic-tags-part3.html**
2. **HTML Requirements**
 - Create the simple document structure using the following tags:
 - **details**
 - **summary**

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/details>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/summary>

10.Semantic Tags – Figure, Figcaption

Create an HTML Document and follow the requirements listed below to complete the task.



Fluffy Kitten

Requirements

1. **File Setup:**
 - Create an **HTML** document named **semantic-tags-part4.html**
2. **HTML Requirements**
 - Create the simple document structure using the following tags:
 - **figure**
 - **figcaption**

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/figure>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/figcaption>

11.Semantic Tags – Nav

Create an HTML Document and follow the requirements listed below to complete the task.

- [Home](#)
- [Products](#)
- [Services](#)
- [Contacts](#)
- [About](#)

Requirements

1. **File Setup:**

- Create an **HTML** document named **semantic-tags-nav.html**
- 2. HTML Requirements**
- Create the simple document structure using the following tags:
 - **nav**

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/nav>

12.Semantic Tags – Time

Create an HTML Document and follow the requirements listed below to complete the task.

The <time> Element Showcase

Examples of the <time> Element

Date Only (YYYY-MM-DD)

The following is a date: January 1, 2023.

Time Only (24-Hour Format)

The following is a time: 13:45 (1:45 PM).

Date and Time Combination

The following is a date and time: January 1, 2023, at 13:45.

Datetime with Timezone

The following is a datetime with a timezone: January 1, 2023, at 13:45 UTC.

Human-readable Date with Datetime Attribute

The following is a human-readable date: New Year's Day 2023.

Requirements

- **File Setup:**
 - Create an **HTML** document named **semantic-tags-time.html**
- **HTML Requirements**
 - Create the simple document structure using the following tags:
 - **time**
- Display at least five different examples of the **<time>** element, including:
 - A date in the format "YYYY-MM-DD".
 - A time in the 24-hour format "HH:MM".
 - A date and time combination.
 - Each example should be accompanied by a brief description of what it represents, utilizing **<p>** elements for descriptions.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/time>

13.Semantic Tags – Blockquote

Create an HTML Document and follow the requirements listed below to complete the task.

Lorem ipsum dolor sit amet consectetur adipisicing elit. Accusamus exercitationem nam obcaecati dolore. Illum, minima. Iusto commodi dolor perspiciatis voluptatum blanditiis suscipit aliquid. Quae, tempora hic fuga maxime iusto ut.

Lorem ipsum dolor sit amet consectetur adipisicing elit. Accusamus exercitationem nam obcaecati dolore. Illum, minima. Iusto commodi dolor perspiciatis voluptatum blanditiis suscipit aliquid. Quae, tempora hic fuga maxime iusto ut.

Lorem ipsum dolor sit amet consectetur adipisicing elit. Accusamus exercitationem nam obcaecati dolore. Illum, minima. Iusto commodi dolor perspiciatis voluptatum blanditiis suscipit aliquid. Quae, tempora hic fuga maxime iusto ut.

Lorem ipsum dolor sit amet consectetur adipisicing elit. Accusamus exercitationem nam obcaecati dolore. Illum, minima. Iusto commodi dolor perspiciatis voluptatum blanditiis suscipit aliquid. Quae, tempora hic fuga maxime iusto ut.

Requirements

- **File Setup:**
 - Create an **HTML** document named **semantic-tags-blockquote.html**
- **HTML Requirements**
 - Create the simple document structure using the following tags:
 - **blockquote**

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/blockquote>

14. Wiki Article

Create an HTML Document and follow the requirements listed below to complete the task.

Semantic HTML: What It Is and How to Use It Correctly

Introduction

Semantic HTML plays a crucial role in web development, providing meaning to the web content beyond the presentation layer. This article explores the core benefits of using semantic elements.

Improved Accessibility

By using elements like <header>, <nav>, and <footer>, developers can create more accessible websites for people using screen readers and other assistive technologies.

SEO Benefits

Semantic markup improves the SEO of web pages by allowing search engines to better understand the content structure, leading to potentially higher rankings in search results.

Ease of Maintenance

Using semantic elements makes it easier for developers to read and maintain code, as the structure of the document is clearer and more intuitive.

Did You Know?

The <article> element can be used independently of the rest of the web page content, making it ideal for blog posts, news articles, and more.

For full screenshot check the screenshot.png file in the folder for the task

Requirements:

- **File Setup:**
 - Create an HTML document named **wiki-article.html**

- **HTML Requirements**

- Create the simple document structure using the following tags:
 - article
 - section
 - h2
 - p
- Use the **<article>** tag to define the main content of your article.
- Break your article into sections using the **<section>** tag, each with a heading **<h2>**
- Ensure the main heading of your article is within an **<h1>** tag, placed inside the **<article>** tag.
- Use **<p>** tags for paragraphs.

Resources

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/article>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/header>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/time>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/section>
- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/p>
- https://developer.mozilla.org/en-US/docs/Web/HTML/Element/Heading_Elements